

Mid-Term Examination for Thermodynamics and Statistical Mechanics 2025-2026(2)

司徒和丞相 20260711

2026.07.11

Problem 1 A system consists of N **distinguishable, independent particles** obeying **Boltzmann statistics**. Each particle can occupy one of three non-degenerate energy levels: $-E$, 0 , and E . The system is in thermal contact with a heat reservoir at temperature T .

Define $\beta = \frac{1}{k_B T}$. The single-particle partition function is:

$$z = e^{\beta E} + 1 + e^{-\beta E} = 1 + 2 \cosh(\beta E)$$

The occupation probabilities for the three levels are:

$$P(-E) = \frac{e^{\beta E}}{z}, \quad P(0) = \frac{1}{z}, \quad P(E) = \frac{e^{-\beta E}}{z}$$

- (a) Find the entropy at $T = 0$ K.
- (b) Find the maximum entropy.
- (c) Find the minimum energy of the system.
- (d) Find the total partition function Z for the N -particle system.
- (e) Find the average energy $\langle U \rangle$ of the system.
- (f) Evaluate the integral

$$\int_0^\infty \frac{C(T)}{T} dT$$

Problem 2 Consider a collection of N two-level systems in thermal equilibrium at temperature T . Each system has only two energy levels: a ground state with energy 0 and an excited state with energy ε . Find and plot vs. temperature:

- (a) The probability that a subsystem is in the excited state
- (b) The entropy of the entire collection

Problem 3

By using Maxwell relations, prove:

$$\left(\partial \frac{C_p}{\partial p} \right)_T = -T \left(\partial^2 \frac{V}{\partial T^2} \right)_p$$

Given hint:

$$\left(\partial \frac{S}{\partial p} \right)_T = - \left(\partial \frac{V}{\partial T} \right)_p$$

Problem 4

Problem 4: Average Energy in the Classical Canonical Ensemble

A particle obeys the classical canonical distribution with energy:

$$\varepsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + ax^2 + bx$$

where a and b are constants. Find the average energy of the particle.

Problem 5

Problem 5: Fraction of Molecules Below the Most Probable Speed

Suppose the number of molecules in a classical ideal gas whose speed is less than the most probable speed v_p is N_p , and the total number of gas molecules is N . Prove that the ratio $r \triangleq \frac{N_p}{N}$ is *independent of the temperature* of the gas.

Problem 6

Problem 6: Fluctuations in Internal Energy and Heat Capacity

Derive that the fluctuations in internal energy at thermal equilibrium are proportional to the heat capacity, and explain why these fluctuations can be neglected under such conditions.

